

ITCS241 Database Management System

Project 01

SoundCloud

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History of SoundCloud and Goals

SoundCloud was founded in 2007 by Alexander Ljung and Eric Wahlforss. The idea to create SoundCloud began when they were both studying at KTH Royal Institute of Technology in Sweden. Then, it has grown to be one of the most extensive music streaming services. There is both free and paid membership, and it is available for desktop and mobile devices. There are many features such as uploading, promoting, sharing, and streaming audio. It has influenced the music industry because of its success. Many artists have used it to launch or advance their music, and many investors support it.

History

- In August 2007, SoundCloud was established in Berlin by Alexander Ljung, Swedish sound designer, and Eric Wahlforss, a Swedish electronic musician.
- In October 2008, the website was launched. Its first purpose is to allow musicians to collaborate by facilitating the sharing and discussion of recordings. It has become a publishing tool for music sharing and began to be a platform for musicians to challenge the dominance of Myspace.
- In April 2009, Doughty Hanson Technology Ventures funded €2.5 million as a Series A to SoundCloud.
- In May 2010, There were one million users.
- In January 2011, it had raised US\$10 million Series B funding from Union Square Ventures and Index Ventures.
- In June 2011, There were five million registered users and investments
- In January 2012, There were 10 million registered users.
- In May 2012, there were 15 million users, and site usage increased by 1.5 million users per month.
- In December 2012, A new APN was released, improving mobile support and adding features such as concurrent listening, site navigation, and creating personal collections. Then, there were 180 million users per month.
- In March 2014, Twitter partnered with SoundCloud for developing its first integrated music app, but the attempt failed.
- In July 2013, There were 40 million registered users and new 20 million users joined per month.

- In January 2014, it announced that it had commenced licensing negotiations with major music companies.
- In May 2015, Twitter was considering the acquisition of SoundCloud. However, the prospect of the purchase was discounted by the media.
- In September 2016, Spotify announced to buy SoundCloud, but after that, they did not.
- In Spring 2017, SoundCloud was sold after not raising the \$100 million needed to support the platform.
- In July 2017, it announced layoffs and the closure of two of its five offices.
- In August 2017, it announced that it reached an agreement on the Raine Group and Temasek investment. Kerry Trainor, a veteran digital media operator, joined the SoundCloud team as Chief Executive Officer, and Michael Weissman joined the SoundCloud team as Chief Operating Officer. Alexander Ljung came to be Chairman of the Board. Eric Wahlforss came to be a Chief Product Officer, and then he transitioned into an advisory role.
- In March 2021, it announced a new pay model for artists, entitled "fan-powered royalties". The income is from the subscription and advertising revenue.

Platforms

- Mobile

There are two mobile apps;

- SoundCloud, which focus on the listener side, such as streaming, searching, playlist and sharing features
- SoundCloud Pulse, which focuses on the uploader side, such as uploading, replying comments, looking for statistics.

SoundCloud Pulse had over 100 million downloads by the end of 2016. In April 2017, the main SoundCloud app for iOS was added to the Chromecast.

- Desktop

In January 2011, SoundCloud released the "SoundCloud Desktop app for Mac", but it was discontinued because of lack of resources to maintain their new desktop application, the mobile applications, and the web browser.

In May 2017, it released an application for Xbox One. Unfortunately, it missed a few features, such as the ability to cast a song from an external device and the ability to go forward or backwards in a playlist without using Cortana.

In the same month, they also launched the "SoundCloud for Windows (Beta)" for Windows 10 in Microsoft Store. It solved the previous problems, but most of their core features such as “repeat” or basic animation are not available. Since then, the app (as 2020) still has not been updated or changed.

Goals

Their goal is to enable artists and listeners to get the most out of the world's top social sound platform by establishing the SoundCloud forum for go, quick, and accessible online assistance channel for solving problems and sharing experiences.

Business Overview

What is SoundCloud?

SoundCloud is the largest audio discovery and artists community platform and services that supports the independence of artists with over 250 million tracks from 30 artists from 190 countries.

For the artists. Artists can upload audios and share them with artists' social networks. They can use this platform to promote the new coming tracks, engage with fanclubs and monetize from the provided tracks.

For the listeners. Listeners can search and find any new audios and share the discoveries into their social networks. They can contribute to the artists they favor by commenting, sharing and liking their audio tracks. They can even save unlimited tracks and playlists for the monthly-subscription model.

In the subscription model, there are 3 choices for users.

SoundCloud Basic	SoundCloud Pro Unlimited	Repost by SoundCloud
Share your sound	Grow your audience	Get heard everywhere
Free	\$12.00 \$8.40 /month when billed yearly	\$2.50 /month billed yearly
Sign up for Free	Try Pro Unlimited	Try Repost
✓ Upload up to 3 hours of your work ✓ Get lossless HD storage ✓ Access basic insights	Everything in Basic & Repost, plus: ✓ Unlock unlimited upload time** ✓ Get paid fairly for your plays ✓ Access advanced audience insights ✓ Replace your track without losing its stats ✓ Pin your favorite tracks	✓ Distribute tracks to all major music services ✓ Extend your reach with promotional tools ✓ Get premium services to help you get paid ✓ Split payments with your collaborators

Basic (Free)

- Free user, they can upload audios up to 3 hours
- Lossless HD audio storage for unlimited
- Be able to get some basic insights

Pro Unlimited (\$8.40 / month)

- Unlimited upload time
- Be able to monetize by users' audio
- Be able to get advanced audience insights
- Upload the overlapped audio track without losing its stats
- Mark users' favorite tracks

Repost (\$2.50 / month)

- Can provide tracks to all popular music services
- Promotional tools accessibility
- Unlock the premium services and monetize
- Divide payments transaction with users' friends

Business rules and Constraints

No.	Rule details	Type of relation
1	One user may upload many tracks. One track belongs to only one user. (unique instance for each user)	1 : M
2	Any user may follow many other users.	M : N
3	One track can be stored in many albums. One album contains many tracks.	M : N
4	One user can upload many albums. One album belongs to one user.	1 : M
5	One track may have many tags. One tag may have many tracks.	M : N
6	One track may have many comments. One comment can be in only one track.	1 : M
7	One user may type many comments. One comment belongs to only one user.	1 : M
8	One user can like many tracks. One track can be liked by many users.	M : N
9	One user can repost many tracks. One track can be reposted by many users.	M : N
10	One user can send many private messages (to other users). Each private message is associated with only one pair of users.	1 : M
11	(History) One user can listen to many tracks. One track can be listened to by many users.	M : N
12	One track can have many users who donate (donor). One donor can contribute many tracks.	M : N

13	One user can have many invoices. One invoice belonged to only one user. (payment transaction log)	1 : M
14	(Subscribe to pro) One invoice can specify only one subscription type One subscription type can be in many invoices	1 : M
15	(Donation) One user (donor) can have many invoices. One invoice belonged to only one donor.	1 : M
16	One invoice can specify only one donation payment One donation payment can be in many invoices	1 : M
17	One playlist can have many tracks. One track can be in many playlists.	M : N
18	One user can have many playlists. One playlist belongs to one user.	1 : M
19	One user may have many notifications. One notification belongs to many users.	M : N
20	One user calls the problems to many operators. One call operator operates many users.	M : N
21	One user can have one user setting. One user setting belongs to only one user.	1 : 1
22	One track may be blacklisted in many countries. One country may blacklist many tracks.	M : N
23	One user can block many users. one user can be blocked by many users. (User <-> User)	M : N
24	One user follows many stations.	M : N

	One station can be followed by many users.	
25	One user can report many tracks. One track can be reported by many users.	M : N
26	One user may have many transaction details. One transaction detail belongs to only one user.	1 : M
27	One user may search many times. One search history belongs to only one user.	1 : M
28	One playlist may have many tags. One tag may be in many playlists.	M : N
29	One user's post may have many tracks. One track can be in many users' posts.	M : N
30	One track has only one license. One license associate with many tracks.	1 : M
31	One track can be created by many users. One user can create many tracks.	M : N
32	One user can edit many tracks. One track can be edited by only many users.	M : N
33	One user can buy a subscription one time per month.	constraint
34	The user cannot follow him/herself.	constraint
35	One unique Email has to belong to only one user.	constraint
36	Only Pro users can monetize their works.	constraint
37	Only Pro users can make unlimited upload time (longer than 3 hours).	constraint
38	Only Pro users can access advanced insight.	constraint
39	Only Pro users can give access to download their tracks more than 100 times.	constraint

40	Only Pro users can get responses when they ask for customer services on that day.	constraint
41	Only Pro users can edit tracks (without creating a new post).	constraint
42	Only Pro users can split payment among collaborators.	constraint

Extended Entity Relational Diagram (EERD)

Entity

1. User

- USER_ID
- NAME
- BIRTH_DATE
- PROFILE_IMG
- HEADER_IMG
- USER_GENDER
- REGISTER_DATE
- REGISTER_TIME
- USER_EMAIL
- HASHED_PASSWORD
- ADDRESS
- DESCRIPTION

2. Track

- TRACK_ID
- TRACK_NAME
- TRACK_LINK
- UPLOAD_DATE
- UPLOAD_TIME
- TRACK_IMG
- DESCRIPTION
- RELEASE_DATE
- RELEASE_TIME
- *TRACK_LISTENED (DERIVED_BY_USER<LISTEN TO>TRACK)*
- *TRACK_LIKED (DERIVED_BY_USER<LIKE>TRACK)*
- *TRACK_COMMENT (DERIVED_BY_USER<COMMENT>TRACK)*

3. Album

- ALBUM_ID
- ALBUM_NAME

- CREATED_DATE
- CREATED_TIME
- ALBUM_IMG
- DESCRIPTION
- ALBUM_LINK
- RELEASE_DATE
- RELEASE_TIME
- *ALBUM_LISTENED (DERIVED_BY_USER<LISTEN TO>TRACK)*
- *ALBUM_LIKED (DERIVED_BY_USER<LIKE>TRACK)*
- *ALBUM_TRACK (DERIVED_BY_ALBUM<CONTAIN>TRACK)*
- *ALBUM_DURATION (DERIVED_BY_ALBUM<CONTAIN>TRACK)*
- *ALBUM_COUNT (DERIVED_BY_ALBUM<CONTAIN>TRACK)*

4. Tag

- TAG_ID
- TAG_NAME

5. Post

- POST_ID
- POST_DETAIL
- POST_DATE
- POST_TIME

6. Payment transaction log

- REF_ID
- LOG_DATE
- LOG_TIME
- RECEIVER_ID
- AMOUNT
- PAYMENT_TYPE

6.1. E-Bank

- BANK_NAME
- BANK_ACCOUNT

6.2. Credit Card

- CARD_NUM
- CARD_TYPE
- CARD_NAME

- CARD_EXP

7. Playlist

- PLAYLIST_ID
- PLAYLIST_NAME
- PLAYLIST_IMG
- PLAYLIST_LINK
- CREATED_DATE
- CREATED_TIME

8. Notification

- NOTIFICATION_ID
- NOTIFICATION_CONTENT
- NOTIFICATION_DATE
- NOTIFICATION_TIME

9. Operator

- OPERATOR_ID
- OPERATOR_NAME
- OPERATOR_IMG
- OPERATOR_EMAIL
- OPERATOR_STATUS
- OPERATOR_HEADER_IMG
- OPERATOR_COUNTRY

10. Setting

- USER_ID
- SETTING_ACCOUNT (PC)
- SETTING_CONTENT (PC)
- SETTING_NOTIFICATIONS (PC)
- SETTING_PRIVACY (PC)
- SETTING_ADVERTISING (PC)
- SETTING_INTERFACE (MOBILE)
- SETTING_ANALYTICS (MOBILE)
- SETTING_COMMUNICATIONS (MOBILE)
- LAST_EDITED_DATE
- LAST_EDITED_TIME

11. Station

- STATION_ID
- STATION_NAME
- STATION_IMG
- *STATION_LENGTH (DERIVED_FROM_TRACK)*
- *STATION_DURATION (DERIVED_FROM_TRACK)*

12. Search_History

- USER_ID
- HISTORY_DATE
- HISTORY_TIME
- HISTORY_CONTENT

13. License

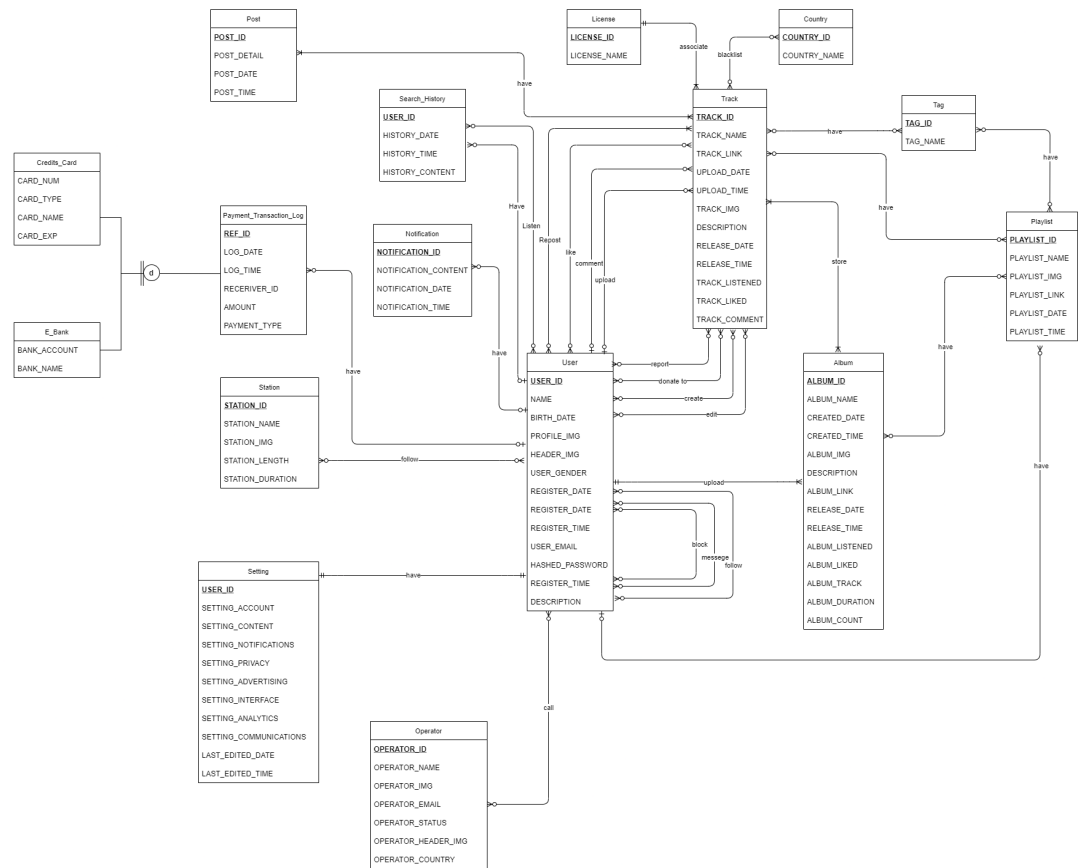
- LICENSE_ID
- LICENSE_TYPE

14. Country

- COUNTRY_ID
- COUNTRY_NAME

Relation

- 1 User <upload> M track
- M User <follow> N User
- M Track <stored> N Album
- 1 User <upload> M Album
- M Track <have> N Tag
- 1 Track <commented> M User
- M User <like> N Track
- M User <repost> N Track
- M User <message> N User
- M User <listen to> N Track (Track history)
- M User <donate to> N Track
- M Playlist <have> N Track
- 1 User <have> M Playlist
- 1 User <have> M Notification
- M User <call> N Operator
- 1 User <have> 1 Setting
- M Track <blacklisted on> N Country
- M User <Block> N User
- M User <Follow> N Station
- M User <report> N Track
- 1 User <have> M History
- M Playlist <have> N Tag
- 1 License <associate> M Track
- M User <create> N Track
- M User <edit> N Track
- 1 User <have> M Payment transaction



Citation

History

Insider. (Aug 27, 2015). The fabulous life of SoundCloud cofounder and CEO Alexander Ljung, from <https://www.businessinsider.com/life-of-soundcloud-founder-ceo-alexander-ljung-2015-8#in-2014-soundcloud-caught-the-attention-of-twitter-which-started-making-a-plan-to-buy-ljungs-company-twitter-had-already-tried-to-launch-a-music-product-to-get-people-to-stay-longer-on-the-platform-but-the-attempt-failed-7>

Wikipedia. (n.d.). SoundCloud. Retrieved September 8, 2021, from <https://en.wikipedia.org/wiki/SoundCloud>

Goals (Vision)

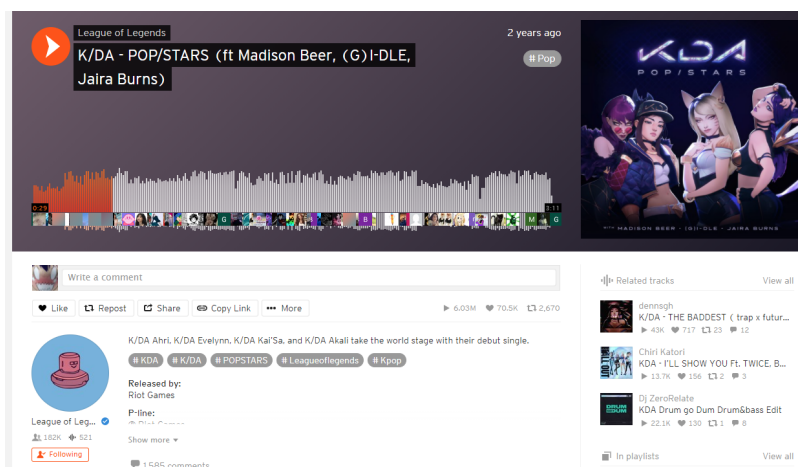
Insided. (n.d.). Case study: SoundCloud, Retrieved September 8, 2021, from [Case study: SoundCloud - inSidedhttps://www.insided.com › case-study-soundcloud-short](https://www.insided.com/case-study-soundcloud-short)

Business Overview

Soundcloud. (n.d.). What is SoundCloud?. Retrieved September 8, 2021, from <https://help.soundcloud.com/hc/en-us/articles/115003570488-What-is-SoundCloud->

Soundcloud. (n.d.). SoundCloud Subscription details. Retrieved September 8, 2021, from <https://checkout.soundcloud.com/pro>

Picture identified the business rules



SoundCloud
Basic

Share your sound

Free

[Sign up for Free](#)

- ✓ Upload up to 3 hours of your work
- ✓ Get lossless HD storage
- ✓ Access basic insights

Limited Time Offer*

SoundCloud
Pro Unlimited

Grow your audience

~~\$12.00~~ **\$8.40/month**
when billed yearly

[Try Pro Unlimited](#)

Everything in Basic & Repost, plus:

- ✓ Unlock unlimited upload time**
- ✓ Get paid fairly for your plays
- ✓ Access advanced audience insights
- ✓ Replace your track without losing its stats
- ✓ Pin your favorite tracks

Repost
by SoundCloud

Get heard everywhere

\$2.50/month
billed yearly

[Try Repost](#)

- ✓ Distribute tracks to all major music services
- ✓ Extend your reach with promotional tools
- ✓ Get premium services to help you get paid
- ✓ Split payments with your collaborators

Get paid: monetize your work			
Make money			
Monetize plays on SoundCloud & other music services ?	✗	✓	✓
Get paid fairly with fan-powered royalties ?	✗	✓	✓
Distribute unlimited releases to all major music services ?	✗	✓	✓
Split payment with your collaborators ?	✗	✓	✓
Choose your monetizable countries ?	✗	✓	✓
Get fan support			
Accept fan donations ?	✓	✓	✗
Submit for playlists			
Pitch your tracks to top playlists across major music services ?	✗	✓	✓

Schemas	<
OAuthToken >	
TrackMetadataRequest >	
CreateUpdatePlaylistRequest >	
TrackDataRequest >	
Found >	
Error >	
TooManyRequests >	
User >	
Me >	
Users >	
Track >	
Tracks >	
TracksList >	

Playlist >	
<pre> Playlists { collection next_href string } </pre>	
PlaylistsArray >	
<pre> Activities { description: User's activities. collection { type string created_at string origin { description: Origin. anyOf -> Track > {...} Playlist > {...} } } next_href string future_href string } </pre>	
WebProfiles >	
Comment >	
Comments >	
Streams >	

**League of Legends**
(•) Artist station

51
TRACKS
3:10:01



♥ Liked

↻ Share

⋮ Add to Next up

⋮ Add to playlist

♥ 1,726

1

League of Legends - K/DA - I'LL SHOW YOU ft. TWICE, Bekuh BOOM, Annika Wells

▶ 444K

2

Fall Out Boy - Centuries

▶ 13.1M

3

TheFatRat - TheFatRat - Rise Up

▶ 2.89M

4

NEFFEX - Fight Back 🍌🔥

▶ 13.9M

5

xslayer - Nightcore - Lost Sky - Dreams Pt. II (feat. Sara Skinner)

▶ 93.2K

6

zXm. - NEONI and burnboy - CHAMPION

▶ 45.3K

7

xslayer - Egzod & Maestro Chives - Royalty (ft. Neoni)

▶ 88.6K

8

NEFFEX - Rumors [Available on Spotify and Apple Music 🍌]

▶ 10.5M

9

666 Records - Blacklite District - More Than Ready

▶ 117K

Based on League of Legends

Artists featured

League of Legends ✓
182K 521

Following

Fall Out Boy
24.2K 26

Follow

TheFatRat ✓
430K 77

Follow

Legal - Privacy - Cookie Policy - Cookie Manager -
Imprint - Creator Resources - Blog - Charts -
Language: English (US)

Autoplay related tracks



Interface style

System >

Notifications



Analytics



Communications

